Exploring the Boundary: Discriminative Model-based Parameter Search for Fault Injection

Ju-Hwan Kim and Dong-Guk Han

Kookmin University, Seoul, Republic of Korea

Abstract. Fault Injection (FI) attacks are physical attacks designed to induce specific malfunctions in target devices. The reliable induction of intended faults requires precise tuning of fault parameters. However, existing parameter search strategies typically suffer from an imbalance between exploration and exploitation. This limitation frequently leads to premature convergence to local optima or inadequate investigation of high-potential regions. In this paper, we propose a novel parameter search framework that employs an ensemble of discriminative models to efficiently generate parameter candidates with high success probabilities. Our approach integrates a regression model to explore the boundary between normal and mute verdicts—leveraging the boundary hypothesis—and a classification model to exploit discovered intended fault samples. Furthermore, we introduce the Refining Successive Halving Algorithm (RSHA) to efficiently identify the global optimum among the discovered fault parameters with statistical confidence. Extensive validation across eight scenarios, involving Voltage Glitching (VG) and Electromagnetic Fault Injection (EMFI), demonstrates that our method consistently outperforms state-of-the-art techniques. Specifically, it identifies up to 42.2× more unique intended fault parameters and improves success rates by up to 20.3 percentage points compared to the best-performing baseline.

Keywords:

1 Introduction

Fault Injection (FI) attacks disrupt the nominal operation of target systems to induce exploitable behaviors. Adversaries typically achieve this by manipulating supply voltages or clock signals [KK99], or by injecting electromagnetic (EM) pulses or laser beams [SA03]. Under such perturbations, devices may exhibit data corruption [PQ03], instruction replacement [GZW+25], or instruction skipping [AK97]. These faults can be exploited to extract secret keys from cryptographic algorithms [BS97] or to bypass authentication mechanisms [VOGT21]. While fault exploitation techniques have been extensively studied [SAW25, KGAS25, WMT25], the practical challenge of reliably inducing specific, exploitable faults has received comparatively less attention.

Reliable fault injection is contingent upon the precise calibration of attack parameters. For instance, the success rate of an Electromagnetic Fault Injection (EMFI) attack depends on the configuration of spatial coordinates (x, y), pulse intensity, and timing delay. The combinatorial complexity of this parameter space makes exhaustive search practically infeasible. In realistic scenarios using high-resolution equipment, the search space can be prohibitively large; our EMFI setup, for example, involves approximately 31 trillion distinct parameter combinations, even excluding timing delay. While heuristics like random

E-mail: zzzz2605@kookmin.ac.kr (Ju-Hwan Kim), christa@kookmin.ac.kr (Dong-Guk Han)

Date of this document: 2026-03-17.

search (Monte Carlo) or grid search are common, they are often inefficient at localizing promising parameter regions [CPB+13]. This limitation has driven research into automated parameter optimization strategies.

1.1 Related Work

Carpi et al. [CPB+13] pioneered automated parameter search strategies by proposing three methodologies: a tailored search (FastBoxing), a binary search variant termed AZB (Adaptive Zoom & Bound), and a Genetic Algorithm (GA)-based approach. Their framework is predicated on the *boundary hypothesis*, which posits that parameters inducing `intended faults` lie in a transition zone where the perturbation is neither benign enough to maintain `normal` behavior nor severe enough to cause a `mute` state (system halt). AZB iteratively bisects intervals between neighboring samples with distinct verdict classes to progressively localize this boundary. Although AZB achieved the highest success rates among their proposed methods, GA was identified as a promising direction due to its potential for optimization via parameter tuning.

Subsequent research has predominantly focused on refining GA [PBJC14, MSPB18, BFP19]. These approaches evolve a population over successive generations using operators tailored to the FI domain. For instance, crossover operators generate offspring by interpolating between parents with different verdict classes, thereby steering the population toward the `normal-mute` boundary [PBJC14]. To address the non-deterministic nature of faults near this boundary, fitness functions prioritize individuals that induce `intended faults` or exhibit `changing` behavior (where fixed parameters yield varying results). Memetic Algorithms (MA), which augment GA with local and tabu search, have also been adopted to enhance local refinement [PBBJ15]. However, both GA and MA depend heavily on the quality of the initial population [KFP21, KOFP22]. Furthermore, they are prone to premature convergence, often stagnating in local optima without sufficiently exploring the broader parameter space.

Recently, Machine Learning (ML) techniques have been adopted to improve search efficiency. Werner et al. framed parameter search as an equipment calibration problem, applying hyperparameter optimization techniques such as SMAC (Sequential Model-based Algorithm Configuration) and SHA (Successive Halving Algorithm) [WMP21]. SMAC uses a random forest surrogate model to estimate fault probabilities and select promising candidates, while SHA iteratively discards underperforming candidates. However, a critical limitation of both techniques is their sensitivity to initialization. SHA is constrained to the initial candidate set and cannot explore beyond it; thus, it fails to identify the global optimum if it is not present initially. Similarly, SMAC relies on a random search (pure exploration) phase to construct its initial surrogate model. If this phase fails to capture meaningful `intended fault` regions within a limited budget, the probabilistic model generalizes poorly. Consequently, SMAC may investigate only a small region if an insufficient number of `intended fault` samples are acquired.

Generative models, specifically Conditional Generative Adversarial Networks (CGAN), have also been used to synthesize parameter candidates [KBV24]. A CGAN is trained on a dataset containing `normal` and `mute` samples acquired via a budget-constrained random search. After training, the generator is conditioned on an intermediate label to generate candidates along the inter-class boundary. However, generative models are famously difficult to train and prone to mode collapse. If the training set contains sparse `intended fault` samples, the generator may overfit, merely reproducing known parameters rather than exploring promising new regions.

In summary, existing methods typically prioritize either exploration or exploitation, failing to optimally balance these conflicting objectives. AZB favors broad exploration but lacks mechanisms to leverage discovered faults for guidance. Conversely, GA, MA, and CGAN are prone to premature convergence on local optima. Similarly, SHA and SMAC primarily focus on refining the initial set, favoring exploitation. This imbalance limits their capacity to reliably identify the global optimum.

1.2 Our Contributions

This paper presents a methodology designed to maximize FI success rates by efficiently localizing optimal parameter configurations. We introduce a two-stage framework comprising a parameter search phase for identifying high-potential candidates and a sieving phase for efficient refinement. Our main contributions are summarized as follows:

A. Discriminative model-based parameter search framework. We propose a novel search strategy using an ensemble of two discriminative models to balance exploration and exploitation. The ensemble comprises: (1) a *regression model* that approximates the boundary between `normal` and `mute` regions; and (2) a *classification model* that pinpoints regions with high probabilities of `intended faults`. The framework dynamically balances exploration and exploitation by adapting the ensemble weight based on model performance.

B. Refining Successive Halving Algorithm (RSHA). We introduce RSHA, a refinement-oriented extension of the conventional SHA [WMP21]. Unlike standard SHA, which balances candidate set size and evaluation count for budget optimization, RSHA is engineered to sieve through a promising candidate set to identify the global optimum with statistical confidence.

C. Extensive validation in diverse attack settings. We validated our method across eight scenarios, covering two fault sources (Voltage Glitching and Electromagnetic Fault Injection), two microcontrollers (8-bit and 32-bit), and two objectives (authentication bypass and Differential Fault Analysis). Our method consistently outperformed state-of-the-art techniques, identifying up to 42.2× more distinct `intended fault` parameters and improving success rates by up to 20.3 percentage points.

D. Empirical investigation of elusive fault regions with indistinct boundaries. We experimentally demonstrate that a subset of `intended fault` parameters can manifest in regions with indistinct boundaries, as observed in a specific hardware configuration. While the majority of exploitable faults align with the boundary hypothesis, this finding exposes the limitations of search strategies predicated solely on this assumption. We provide a critical analysis of the trade-off between search efficiency and coverage, offering strategic guidelines for addressing such atypical vulnerability clusters in comprehensive security assessments.

2 Preliminaries

2.1 Notation

Table 1 summarizes the primary notation used in this paper.

2.2 Fault Injection Attacks

FI attacks perturb devices to induce malfunctions like data corruption (e.g., Differential Fault Analysis (DFA) [PQ03]) or modification of the program flow (e.g., authentication bypass [CPB+13]). Reliable induction of such exploitable faults requires precise calibration of the FI parameters.

Table 1: Summary of notation.

Symbol	Description
$\mathcal{P}, \mathcal{V}$	Fault parameter space and set of verdict classes
P, V	Random variables representing fault parameter and verdict class
p, v	Fault parameter vector and verdict value ($p \in \mathcal{P}$, $v \in \mathcal{V}$)
v_{int}	Target verdict class (i.e., `intended fault`)
m	Ensemble of discriminative models estimating $\Pr(V \mid P)$
$m(p)_v$	Predicted probability of verdict v given parameter p
$\mathcal{F}$	Collected fault injection dataset (parameter-verdict pairs)

Fault Parameters. We categorize parameters into two classes based on their adjustability:

- **Hardware parameters:** Static physical configurations, such as wire length for VG, or probe diameter and coil winding direction for EMFI. These are generally fixed and ill-suited for algorithmic optimization.
- **Software parameters:** Dynamic, programmable variables. For VG, these include timing (delay), glitch width, and repeat count. For EMFI, they include timing, spatial coordinates (x, y), and pulse voltage.

This work focuses on the automated optimization of software parameters.

Fault Verdict Classes. The outcome of a fault injection is classified as:

- `normal`: The device maintains nominal operation despite the perturbation.
- `mute`: The device halts, resets, or becomes unresponsive.
- `intended fault`: The specific, desired fault occurs (e.g., a single-byte corruption facilitating DFA).
- `inconclusive`: Anomalous behavior distinct from the target fault (e.g., unexploitable corruption).

Additionally, certain prior studies incorporate the notion of a `changing` verdict to describe parameters that yield non-deterministic outcomes across multiple trials with identical settings.

Boundary Hypothesis. As discussed in Section 1.1, most FI parameter search methods [CPB+13, PBJC14, KBV24] build upon the *boundary hypothesis* proposed by Carpi et al. This hypothesis postulates that the parameter region containing `intended faults` lies within a transition zone, where the perturbation is neither benign enough to preserve `normal` behavior nor severe enough to trigger a `mute` state. Specifically, a `normal` verdict implies insufficient intensity or EM probe placement over a component irrelevant to the target operation, whereas a `mute` verdict implies excessive intensity or placement over a sensitive component causing system failure. The `intended fault` manifests in the delicate margin between these extremes. Consequently, optimal parameters are predominantly localized along the boundary separating the `normal` and `mute` regions.

2.3 Discriminative and Generative Models

In machine learning, supervised learning paradigms aim to learn a mapping between input features $x \in \mathcal{X}$ and corresponding labels $y \in \mathcal{Y}$. From a probabilistic perspective, these models are broadly categorized into discriminative and generative models.

A discriminative model estimates the conditional probability $\Pr(Y \mid X)$, directly modeling the decision boundary between classes in the feature space. Conversely, a generative model learns the joint probability distribution $\Pr(X, Y)$ or the class-conditional density $\Pr(X \mid Y)$. For instance, Conditional Generative Adversarial Networks (CGANs) model $\Pr(X \mid Y)$ to synthesize new samples for a target class. However, generative models are notoriously difficult to train. Traditional statistical models (e.g., Naive Bayes) often converge to higher asymptotic errors compared to their discriminative counterparts [NJ01]. Furthermore, modern deep generative models, such as GANs, are prone to training instability and mode collapse—a failure mode where the generator produces a limited variety of samples, failing to capture the full diversity of the underlying data distribution [GPM+14, SC22].

The relationship between these paradigms is governed by Bayes' rule:

$$\Pr(Y \mid X) = \frac{\Pr(X \mid Y)\Pr(Y)}{\Pr(X)}.$$

If the joint distribution $\Pr(X, Y)$ is known, the conditional probability $\Pr(Y \mid X)$ can be derived by marginalizing over $\mathcal{X}$. Thus, generative models can function as discriminative models. The converse, however, typically does not hold. For general real-world data, estimating the marginal likelihood $\Pr(X)$ is often intractable due to the manifold hypothesis [BCV13]. Consider the MNIST dataset, consisting of 28×28 grayscale images of handwritten digits, as an illustrative example. While the cardinality of the input feature space is exponentially large ($2^{28\times 28} \approx 10^{236}$), valid images reside on a lower-dimensional manifold occupying a negligible volume of the ambient space. Due to this inherent difficulty in defining $\Pr(X)$ for real-world data, discriminative models generally cannot function as generative models.

3 Methodology

3.1 Theoretical Basis

As delineated in Section 2.3, discriminative models generally cannot function as generative models due to the intractability of estimating the marginal distribution $\Pr(X)$ in general settings. However, the FI parameter space exhibits a unique characteristic that we exploit to overcome this limitation. Let the domains of the FI parameters be denoted by $\mathcal{P}_1, \mathcal{P}_2, \ldots, \mathcal{P}_n$. For instance, in an EMFI attack, these domains may correspond to pulse intensity, timing delay, and spatial coordinates. The complete parameter space $\mathcal{P}$ is defined as the Cartesian product: $\mathcal{P} = \mathcal{P}_1 \times \mathcal{P}_2 \times \cdots \times \mathcal{P}_n$. Unlike natural data manifolds, every vector p within this Cartesian product constitutes a valid FI configuration. Given that the valid parameter space corresponds exactly to the entire Cartesian product, we assume an uninformative prior over $\mathcal{P}$. This implies a uniform distribution, such that $\Pr(P = p) = 1/|\mathcal{P}|$ for all $p \in \mathcal{P}$. We formalize this observation in Property 1.

Property 1. Let P and V denote random variables representing the fault parameter and verdict. For any verdict v, the conditional probability $\Pr(V = v \mid P = p)$ is proportional to the posterior $\Pr(P = p \mid V = v)$.

Proof. Applying Bayes' theorem, the posterior probability is expressed as:

$$\Pr(P = p \mid V = v) = \frac{\Pr(V = v \mid P = p)\Pr(P = p)}{\Pr(V = v)}.$$

Since $\Pr(P = p) = 1/|\mathcal{P}|$ is constant, and $\Pr(V = v)$ is independent of p:

$$\Pr(P = p \mid V = v) = \alpha \cdot \Pr(V = v \mid P = p),$$

where $\alpha = 1/(|\mathcal{P}| \cdot \Pr(V = v))$ is a constant. □

This implies that within the FI parameter space, a discriminative model serves as an effective surrogate scoring function for a generative model. Specifically, let m be a discriminative model approximating the posterior distribution over verdict classes: $m(p)_v \approx \Pr(V = v \mid P = p)$. According to Property 1, the output $m(p)_{v_{\text{int}}}$ provides a value proportional to the likelihood $\Pr(P = p \mid V = v_{\text{int}})$, where v_{int} denotes the `intended fault` verdict class.

Leveraging this proportionality, we employ rejection sampling to generate parameter candidates likely to induce the `intended fault`, bypassing the need for the normalization constant α. Rejection sampling is a fundamental technique for sampling from a complex target distribution using a simpler proposal distribution [Bis07]. The method involves drawing samples from a simpler proposal distribution, denoted as $\Pr(Q = p)$, and stochastically accepting or rejecting them. To generate a sample from the target distribution $\Pr(P \mid V = v_{\text{int}})$, the standard procedure is defined as follows:

1. Determine a constant M such that $\Pr(P = p \mid V = v_{\text{int}}) \leq M \cdot \Pr(Q = p)$ for all $p \in \mathcal{P}$.

2. Draw a sample p from the proposal distribution Q and a sample u from the uniform distribution $\mathcal{U}(0, 1)$.

3. Accept the sample p if it satisfies the following criterion; otherwise, reject it:

$$u < \frac{\Pr(P = p \mid V = v_{\text{int}})}{M \cdot \Pr(Q = p)}. \tag{1}$$

Property 1 enables a streamlined rejection sampling procedure to generate promising parameter candidates via the discriminative model m.

Property 2. By adopting a uniform distribution over $\mathcal{P}$ as the proposal distribution, the acceptance criterion for rejection sampling in Equation 1 simplifies to $u < m(p)_{v_{\text{int}}}$.

Proof. Recall from Property 1 that the target posterior is proportional to the likelihood: $\Pr(P = p \mid V = v_{\text{int}}) = \alpha \cdot \Pr(V = v_{\text{int}} \mid P = p)$. To streamline the sampling process, we select a uniform proposal distribution, $\Pr(Q = p) = 1/|\mathcal{P}|$, and define the scaling constant as $M = \alpha \cdot |\mathcal{P}|$. This choice satisfies the envelope condition since probabilities are bounded by 1. Substituting these terms into Equation 1, we obtain:

$$u < \frac{\alpha \cdot \Pr(V = v_{\text{int}} \mid P = p)}{M \cdot \Pr(Q = p)} = \Pr(V = v_{\text{int}} \mid P = p).$$

Consequently, by approximating the likelihood with the discriminative model m, the criterion simplifies to verifying whether $u < m(p)_{v_{\text{int}}}$. □

3.2 Discriminative Model-based Parameter Search

We propose a framework that leverages rejection sampling with discriminative models to generate promising parameter candidates. Consistent with the boundary hypothesis, we design an ensemble comprising a regression model specialized for exploring the boundary and a classification model dedicated to exploiting acquired `intended fault` samples. The trade-off between exploration and exploitation is dynamically balanced by the ensemble weights. Figure 1 outlines the framework.

3.2.1 Initial Data Acquisition and Augmentation

The framework commences with a random search phase to establish a preliminary training dataset. The primary objective of this phase is to obtain a coarse characterization of

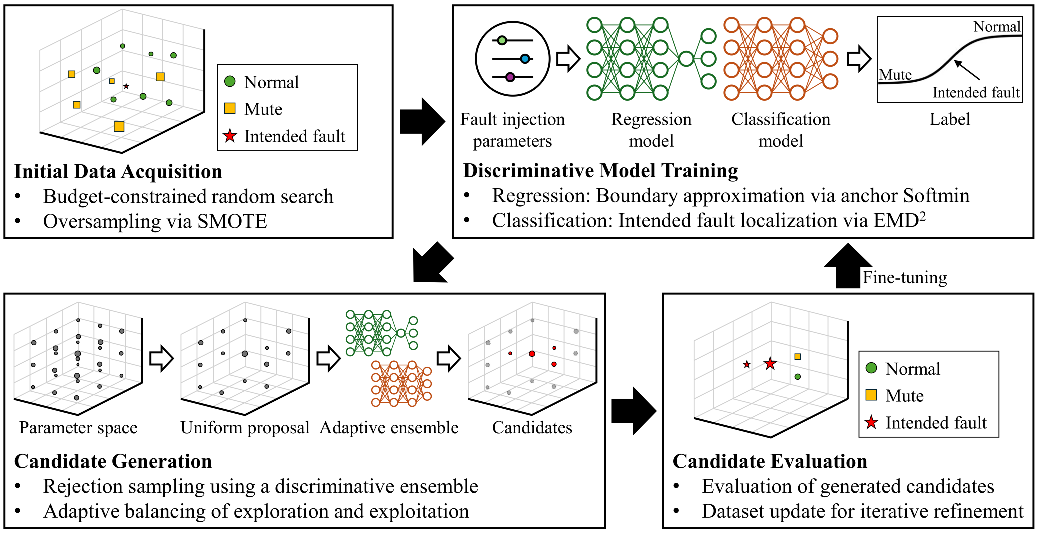


Figure 1: Schematic overview of the proposed framework, illustrating the cycle of candidate generation and model fine-tuning.

the boundary separating the `normal` and `mute` regions, rather than explicitly capturing `intended fault` samples. Previous studies have demonstrated that FI parameters exhibiting distinct verdict classes typically manifest as localized clusters within the parameter space [CPB+13, PBJC14, KBV24]. Consequently, a limited budget of a few thousand injections is generally sufficient to delineate these regions. While the acquisition of `intended fault` samples is advantageous, it is not a strict prerequisite at this stage.

The distribution of fault verdict classes is generally imbalanced, a characteristic that impedes effective model training. To mitigate this imbalance, we employ the Synthetic Minority Oversampling Technique (SMOTE) [CBHK02] to augment minority classes. SMOTE generates synthetic instances by interpolating between existing samples, a strategy particularly well-suited to the continuous topology of the FI parameter space. However, achieving a perfect balance through oversampling alone would necessitate excessive augmentation, thereby increasing the risk of overfitting. To address this, we constrain the augmentation ratio of minority classes to a maximum factor of 10, ensuring that the synthetic population does not overwhelm the majority class. Furthermore, as SMOTE requires a minimum of k neighbors to construct an interpolation set, we apply random oversampling prior to SMOTE if a minority class—specifically the `intended fault` class—contains fewer than $k+1$ samples. To complement this data-level augmentation, we incorporate a cost-sensitive learning strategy by assigning class weights inversely proportional to the sample frequency of each class, normalized to have a unit mean. Our framework focuses exclusively on the three primary verdict classes: `normal`, `mute`, and `intended fault`. Non-deterministic verdicts, such as `changing`, which necessitate repeated injections for classification, are excluded from the framework. Finally, the parameters are min-max normalized to $[-1, 1]$.

3.2.2 Ensemble of Discriminative Models

To leverage the boundary hypothesis, we design two complementary discriminative models: a classification model and a regression model. We cast the problem as an ordinal classification task, mapping the labels 0, 1, and 2 to the `mute`, `intended fault`, and `normal` classes, respectively. Both models are trained to assign high probabilities to the intermediate class (v_{int}) for inputs near the boundary, even in the absence of observed `intended fault` samples. These models are combined in an ensemble to dynamically balance exploration and exploitation.

Classification Model. The classification model is implemented as a standard multi-class classifier with a Softmax output layer, regularized by an ordinal loss function: the Squared Earth Mover's Distance (EMD^2) [HYS16]. Let $\mathbf{o}$ and $\mathbf{y}$ represent the model's output probability distribution and the one-hot encoded ground-truth label, respectively. The EMD^2 regularization is defined as:

$$\mathrm{EMD}^2(\mathbf{o}, \mathbf{y}) = \sum_{i=0}^{2} \big(\mathrm{CDF}_i(\mathbf{o}) - \mathrm{CDF}_i(\mathbf{y})\big)^2,$$

where CDF_i denotes the i-th element of the cumulative distribution function. The final objective function is the sum of the Cross-Entropy (CE) loss and the EMD^2 regularization term. Unlike conventional classifiers optimizing solely for CE, which target only the correct class probability, EMD^2 imposes penalties proportional to the ordinal distance from the ground truth. This ensures that misclassifications into distant classes incur a higher penalty than those into adjacent classes. Consequently, the model infers the boundary structure by implicitly increasing the probability of the intermediate class ($\mathbf{o}_1$) for parameters situated in the ambiguous transition zone between `normal` and `mute` regions. When `intended fault` samples are available, the model prioritizes these observations by assigning higher probabilities in their vicinity. This distinction arises because $\mathbf{o}_1$ is increased indirectly by regularization for boundary parameters, while it is directly maximized by the CE loss for explicit `intended fault` samples. Thus, the classification model favors *exploitation*.

Regression Model. The regression model is architected to explicitly learn the boundary structure. To foster exploration, the model compresses the latent space to a single dimension scalar x, which is subsequently mapped to a probability distribution over the verdict classes. We define three fixed anchors at -1, 0, and 1, corresponding to the `mute`, `intended fault`, and `normal` classes, respectively. Given the output x, we compute the squared Euclidean distances to these anchors to generate a distance vector $\mathbf{z}$, and apply a Softmin function (softmax of negative distances) to derive the final probability vector $\mathbf{o}$:

$$\mathbf{z} = \big[(x-(-1))^2, (x-0)^2, (x-1)^2\big], \quad \mathbf{o}_i = \frac{e^{-\mathbf{z}_i}}{\sum_{j=0}^{2} e^{-\mathbf{z}_j}}.$$

Therefore, the model is optimized so that inputs from each class are embedded near their corresponding anchors via CE loss. Unlike the classification model, where each output element is determined by independent weights, the regression model determines all three elements jointly based on a single variable x. Consequently, even when `intended fault` samples are present, the model tends to distribute high $\mathbf{o}_1$ probability broadly along the boundary. Thus, the regression model is inherently geared towards *exploration*.

Ensemble. To harness the complementary strengths of the two models, we construct an ensemble that effectively balances between exploration and exploitation. The ensemble's predicted probability distribution is a weighted average of the individual model outputs. We determine the optimal weight by minimizing the Brier score [Bri50] on the collected fault results $\mathcal{F}$. To ensure that both models contribute, we evaluate a discretized set of candidate weights $\mathcal{W}$ strictly within the open interval $(0, 1)$ (e.g., $\mathcal{W} = \{0.01, 0.02, \ldots, 0.99\}$). The optimal weight w^* is selected as follows:

$$w^* = \operatorname*{argmin}_{w \in \mathcal{W}} \frac{1}{|\mathcal{F}|} \sum_{(p,v)\in\mathcal{F}} ||(w \cdot m_r(p) + (1-w)\cdot m_c(p)) - \mathrm{onehot}(v)||_2^2\,,$$

where m_c and m_r denote the classification and regression models, respectively.

3.2.3 Parameter Generation via Rejection Sampling and Fine-tuning

Property 2 implies that the acceptance criterion for rejection sampling in Equation 1 can be simplified. Consequently, promising candidates are generated via the following procedure.

1. Draw a sample p from the uniform distribution over the parameter space $\mathcal{P} = \mathcal{P}_1 \times \cdots \times \mathcal{P}_n$ and a sample u from the uniform distribution $\mathcal{U}(0, 1)$.
2. Accept the sample p if it satisfies the following criterion; otherwise, reject it:

$$u < m(p)_{v_{\text{int}}}.$$

Consequently, the discriminative ensemble m is directly utilized for generating parameter candidates with a high probability of inducing the `intended fault`.

The results obtained from fault injections using these candidates are subsequently used to improve the deep learning models. Specifically, the training dataset $\mathcal{F}$ is updated with the newly acquired results, followed by the fine-tuning of both discriminative models and the ensemble weight. This iterative process is critical. In the initial random search phase, the absence or scarcity of `intended fault` samples limits accurate learning of the `intended fault` distribution; the model captures only the boundary. However, by performing injections with the generated promising parameters, we can intensively explore these regions and acquire `intended fault` samples. Consequently, iterative fine-tuning with batches of several thousand parameters allows the ensemble to progressively learn the distribution of `intended faults` with greater precision, ultimately yielding a large set of effective fault parameters.

3.3 Refining Successive Halving Algorithm (RSHA)

While the exploration phase yields a set of effective parameters capable of inducing the `intended fault`, the ultimate objective is to isolate the global optimum that maximizes attack reliability. To achieve this, we introduce the Refining Successive Halving Algorithm (RSHA).

SHA is a hyperparameter optimization technique that iteratively prunes the lower-performing fraction of candidates while allocating increasing resources to the survivors. Unlike standard SHA, which typically seeks an optimal solution within an arbitrarily initialized set of parameters, RSHA is specifically engineered to sieve through a pre-identified set of high-potential candidates to pinpoint the global optimum. Given the stochastic nature of FI attacks, ensuring statistical significance in the evaluation is crucial. We model each fault injection attempt as a Bernoulli trial. To estimate the true success rate s with a confidence level of $1 - \alpha$, the estimation error ϵ is bounded by the sample size n_{eval} as follows:

$$\epsilon = Z_{\alpha/2}\sqrt{\frac{s(1-s)}{n_{\text{eval}}}} \leq \frac{Z_{\alpha/2}}{2\sqrt{n_{\text{eval}}}} \quad (\because s(1-s) \leq 1/4), \tag{2}$$

where $Z_{\alpha/2}$ denotes the critical value of the standard normal distribution. This bound implies that reliably distinguishing the optimal parameter requires a guaranteed cumulative number of evaluations. Consequently, distinct from conventional SHA, RSHA is designed to ensure that the final surviving candidate undergoes a target cumulative number of evaluations, n_{eval}. Furthermore, to mitigate the risk of prematurely discarding potential optima due to high variance in early evaluation stages, we enforce a minimum evaluation count n_{min}.

The procedural details of RSHA are outlined in Algorithm 1. Let η denote the reduction factor. To guarantee that the final survivor reaches n_{eval} trials, the initial trial budget

n_{test} is derived from the geometric series governing the reduction process. Note that, unlike standard approaches, RSHA aggregates the success score continuously rather than evaluating candidate performance independently within each round, thereby minimizing estimation error and ensuring statistical robustness.

Algorithm 1 Refining Successive Halving Algorithm (RSHA)

Require: Fault results $\mathcal{F}$, target cumulative evaluations n_{eval}, minimum evaluations per candidate $n_{\min}$, reduction factor η

Ensure: Optimal parameter p^*

$\mathcal{C} \leftarrow \{p \mid (p, v) \in \mathcal{F} \land v = v_{\text{int}}\}$ ▷ Initialize candidate set
Initialize success score $S(p) \leftarrow 0$ for all $p \in \mathcal{C}$
$r \leftarrow \lceil \log_\eta |\mathcal{C}| \rceil$ ▷ Expected number of reduction rounds
$n_{\text{test}} \leftarrow \max\big(n_{\min}, \lceil n_{\text{eval}} \cdot (\eta - 1)/(\eta^r - 1) \rceil\big)$
$n_{\text{cum}} \leftarrow 0$
while $|\mathcal{C}| > 1$ and $n_{\text{cum}} < n_{\text{eval}}$ **do**
    $n_{\text{cum}} \leftarrow n_{\text{cum}} + n_{\text{test}}$
    for each $p \in \mathcal{C}$ **do** ▷ Inject faults n_{test} times for each parameter
        for $i = 1$ to n_{test} **do**
            $v \leftarrow \text{FaultInjection}(p)$
            if $v = v_{\text{int}}$ **then**
                $S(p) \leftarrow S(p) + 1$
            end if
        end for
    end for
    $n_{\text{test}} \leftarrow \eta \cdot n_{\text{test}}$
    $k \leftarrow \lceil |\mathcal{C}|/\eta \rceil$ ▷ Retain top $1/\eta$ candidates
    $\mathcal{C} \leftarrow$ top-k parameters of $\mathcal{C}$ based on $S(p)$
end while
return $\text{argmax}_{p \in \mathcal{C}} S(p)$

The complexity of Algorithm 1 is contingent upon the candidate set size. Note that the number of fault injections per round is approximately $n_{\text{test}} \cdot |\mathcal{C}|$. In scenarios where the budget permits full execution ($\eta^r - 1 \leq n_{\text{eval}}$), the total number of injection attempts approximates $r \cdot n_{\text{test}} \cdot |\mathcal{C}| = r \cdot \lceil n_{\text{eval}} \cdot (\eta - 1)/(\eta^r - 1) \rceil \cdot |\mathcal{C}|$. Conversely, in constrained scenarios necessitating early termination, the complexity approximates $\lceil \log_\eta (n_{\text{eval}} + 1) \rceil \cdot n_{\min} \cdot |\mathcal{C}|$.

4 Experimental Evaluation

4.1 Experimental Environment

We evaluate the proposed framework across eight distinct fault injection scenarios. Our testbed comprises two physical fault sources—voltage glitching (VG) and Electromagnetic Fault Injection (EMFI)—targeting two commercial-off-the-shelf microcontrollers (MCUs): the 8-bit ATXmega128D4 (AVR XMEGA architecture) [Newb] and the 32-bit SAM4S2A (Arm Cortex-M4 architecture) [Newc]. We define two distinct fault objectives: (1) *authentication bypass*, where the goal is to subvert a PIN verification routine (Listing 1) to return `SUCCESS` despite an incorrect input; and (2) *Differential Fault Analysis (DFA)*, aiming to induce a single-byte corruption of the ninth-round SubBytes output of AES-128, consistent with the fault model proposed by Piret and Quisquater [PQ03].

```c
// ... (variable initialization)
for (i = 0; i < 4; i++) {
    if (pin[i] == input[i])
        digits_ok++;
}
if (digits_ok == 4)
    return SUCCESS;
else
    return FAIL;
```

Listing 1: Target code for PIN verification.

Experimental Setup. For the fault injection equipment, we used the ChipWhisperer-Husky [Newa] for VG and the ChipSHOUTER [Newd] for EMFI. Trigger signals and external offsets were calibrated to synchronize with the execution of the target operation (i.e., the PIN check logic in Listing 1 or the AES ninth-round SubBytes layer). The EMFI spatial search space was defined to cover the entire die surface. Table 2 delineates the configurable parameters and their respective search spaces.

Table 2: Specifications of configurable fault injection parameters and their respective search spaces for the targets.

Source	Parameter	Description	Search Space	Resolution
Timing	External offset	Delay from trigger to glitch injection.	$[0, 72]$ (PIN, 8-bit) $[0, 168]$ (PIN, 32-bit) $[0, 260]$ (AES, 8-bit) $[0, 356]$ (AES, 32-bit)	1 cycle
	Offset	Phase offset from the rising clock edge.	$(-50\%, 50\%)$	1/2800 cycle
VG	Width	Pulse width as a fraction of clock period.	$[0\%, 50\%)$	1/2800 cycle
	Repeat	Number of consecutive glitches per trigger.	$[1, 10]$	1
EMFI	Voltage	Pulse voltage applied to the probe.	$[150, 500]$	1 V
	Position (x, y)	Spatial coordinates of the probe.	$[0, 330\,000]$ (8-bit) $[0, 300\,000]$ (32-bit)	1 step

Implementation Specifics. The proposed framework (Section 3.2) was instantiated with a sampling batch size of $5,000$. The search process was executed for 20 iterations, yielding a total budget of $100,000$ injections per experiment. The discriminative models were instantiated as Multi-Layer Perceptrons (MLPs). Each hidden layer consists of fully connected units followed by batch normalization and Leaky-ReLU activation. We performed a grid search over four configurations, varying the hidden layer architecture (three layers of 128 units versus four layers of 256 units) and learning rates (10^{-3} and 10^{-4}). Training proceeded for 100 epochs using the Adam optimizer with a batch size of 100. For the RSHA phase, the minimum evaluation count was set to $n_{\min} = 2$, the total evaluation budget to $n_{\text{eval}} = 600$, and the reduction factor to $\eta = 3$. To validate the reliability of the optimal parameter identified, we performed $3,000$ independent validation injections. This sample size ensures that the estimation error ϵ is bounded within 2.34% at a 99% confidence level ($\alpha = 0.01$) based on the Equation 2.

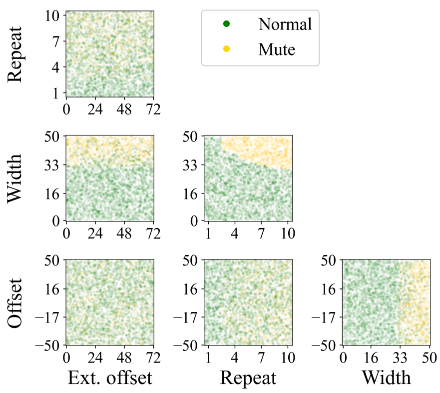
Baseline Methodologies. We benchmark our methodology against seven parameter search strategies, comprising six state-of-the-art techniques and random search. To ensure a fair comparison, a strict budget of $100,000$ injections was enforced across all methods. For AZB [CPB+13], we set the number of injections per parameter to five to enable the detection of `changing` behavior. We evaluated three grid initialization resolutions: 3, 5, and 10 subdivisions per dimension. For the Genetic Algorithm (GA) [PBJC14], we set the tournament size to 3 and the mutation rate to 0.1 as specified in the original paper. Consistent with the AZB methodology, we set the number of injections per candidate evaluation to five. We evaluated population sizes of 30, 100, and 300. The Memetic Algorithm (MA) [PBBJ15] was configured identically to the GA, with local search parameters adapted from the original study. For SHA and SMAC [WMP21], we adopted the reference configurations provided by the authors. Finally, for the CGAN-based approach [KBV24], we aligned the neural network architecture and hyperparameters with those used in our framework, while retaining other settings from the original publication.

4.2 Parameter Search Efficiency

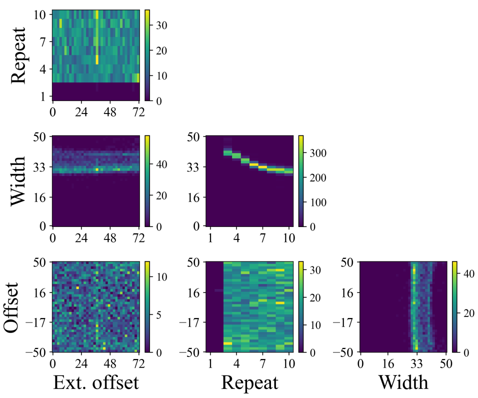
Visualization of the Search Process. Figure 2 illustrates the evolution of the parameter search for the VG scenario targeting PIN verification on the 8-bit MCU. Due to the high dimensionality of the parameter space, samples are projected onto 2D planes for visualization. Figure 2a depicts the initial data acquisition phase, comprising 5,000 random injections. Corroborating observations from prior studies [CPB+13, PBJC14, KBV24], the `normal` and `mute` classes form distinct clusters. This confirms that a limited budget of a few thousand injections is sufficient to coarsely delineate the boundary. Parameters such as offset and external offset exhibited negligible correlation with the separation between `normal` and `mute` regions. In contrast, the boundary is clearly defined by the glitch intensity parameters—specifically width and repeat—as shown in the central plot of Figure 2a. In accordance with the boundary hypothesis, a core postulate of parameter search strategies [CPB+13, PBJC14, PBBJ15, KBV24], `intended faults` are expected to reside within this transition zone.

Figure 2b illustrates the distribution of parameters generated by the proposed method. The model effectively targets the boundary, generating promising candidates for subsequent iterations. Specifically, for parameters unrelated to the boundary, such as offset and external offset, candidates were generated across the entire search space. Conversely, for the width-repeat parameters, which exhibit a clear boundary, candidates were generated in regions consistent with the random search results. Figure 2c presents the distribution following the acquisition of `intended fault` samples. Notably, the generation process dynamically shifts focus to the vicinity of these samples, while still allocating candidates to other boundary regions to maintain exploration. Notably, the generation process dynamically shifts focus to the vicinity of these samples. This dual capability confirms that our framework effectively balances between exploration and exploitation. Finally, Figure 2d displays the results after the full budget of 100,000 injections. The `intended fault` parameters are densely concentrated along the width-repeat boundary. Furthermore, the offset- and external offset-related projections indicate that the `intended fault` distribution forms specific clusters within the broader boundary region. In particular, we observed a correlation between the offset and the repeat/width parameters, a pattern similarly identified in other search techniques. This finding implies that mere boundary exploration is insufficient; active exploitation is requisite to isolate these critical sub-regions.

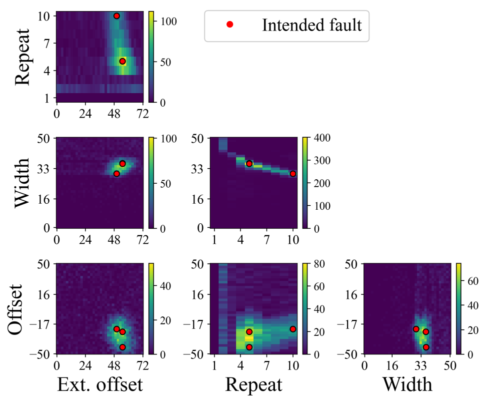
Comparative Analysis of Search Efficacy. Figure 3 compares our method against state-of-the-art techniques for the VG scenario targeting PIN verification on the 8-bit MCU, projecting results onto the width-repeat plane with clear boundary. Each plot visualizes the execution that yielded the highest number of `intended faults` for the respective



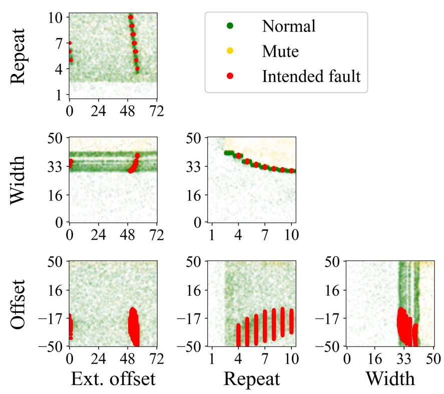
(a) Initial random search results.



(b) Distribution generated by the ensemble trained on the initial dataset.



(c) Distribution after first fine-tuning iteration.



(d) Final cumulative fault results showing the discovered parameter landscape.

Figure 2: Visualization of the parameter search evolution for VG targeting PIN verification on the 8-bit MCU.

technique. As discussed in Section 1.1, AZB exhibits a pronounced bias towards exploration; while it covers a broad area, it fails to sufficiently exploit discovered `intended fault` samples, thereby leaving promising regions unexplored within a limited budget. Conversely, GA, MA, SHA, SMAC, and CGAN demonstrate a marked tendency towards exploitation. Once an `intended fault` sample is identified, these methods focus almost exclusively on its immediate neighborhood. As shown in the figure, GA and MA concentrate their search only in the vicinity of previously found `intended fault` samples. This behavior empirically demonstrates the phenomenon of premature convergence to local optima. SHA allocates more budget to parameters with higher success rates from the initial random set, resulting in the repeated evaluation of a small subset of candidates; consequently, it identified only a single parameter. Similarly, SMAC explores the neighborhood of `intended fault` samples obtained from the initial random search phase. CGAN generated parameters in the boundary region prior to acquiring `intended fault` samples; however, subsequently, it exhibited mode collapse, repeatedly generating only those samples included in the training data. Consequently, significant regions of `intended fault` parameters remain unexplored in these exploitation-oriented methods. Recall that this figure illustrates the result corresponding to the largest number of `intended fault` parameters for each method. In contrast, our method achieves a balance between exploration and exploitation, successfully covering the most extensive `intended fault` region. Moreover, Figure 6 quantifies this performance; our approach identified at least 42× more distinct `intended fault` parameters than the best-performing baseline.

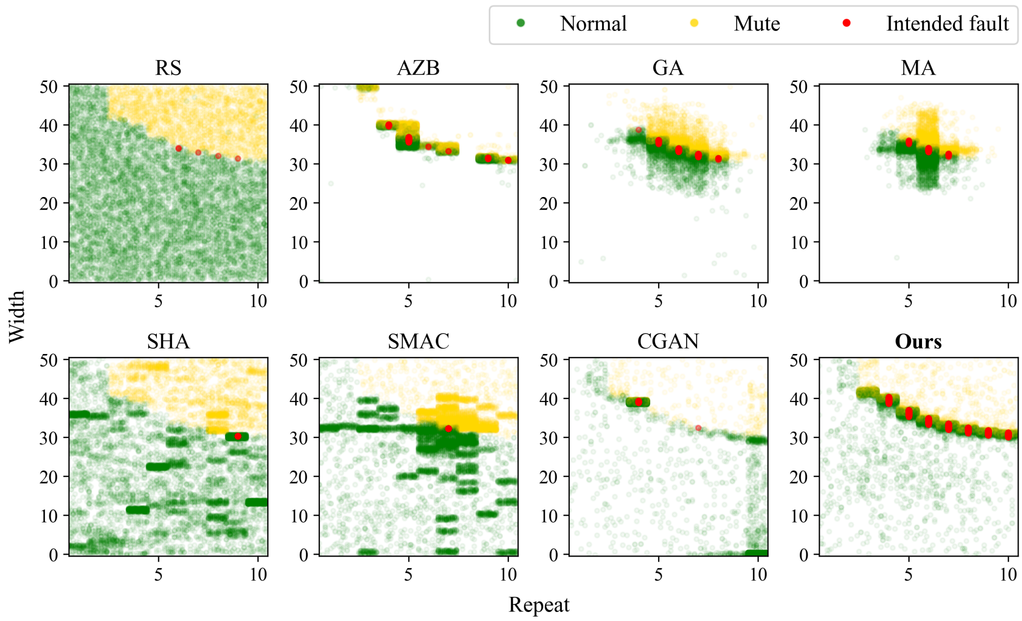
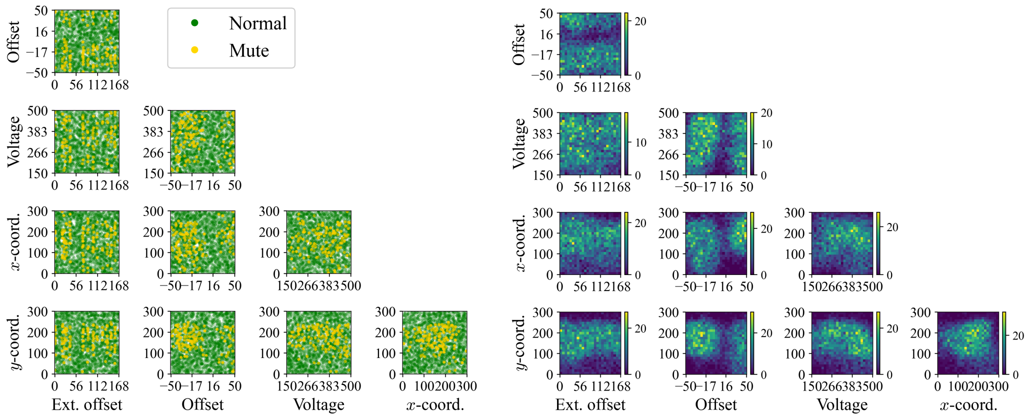


Figure 3: Comparison of candidate parameter distributions for VG (PIN, 8-bit MCU) projected onto the width-repeat plane.

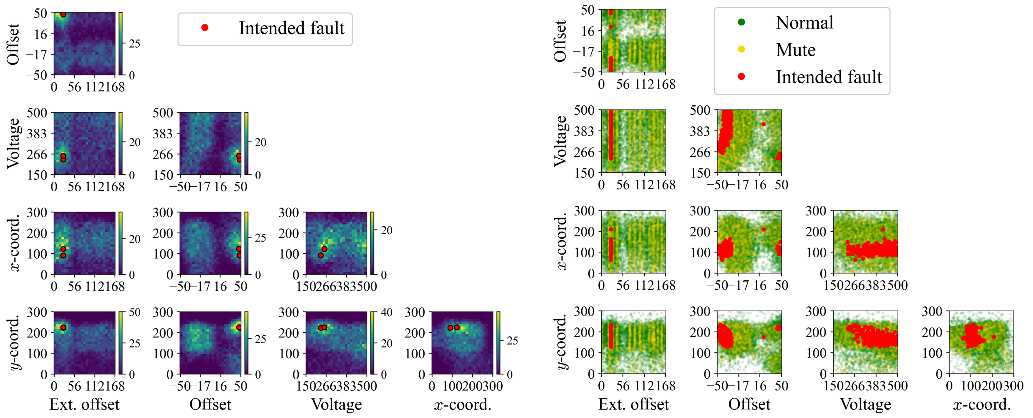
Analysis of the EMFI Scenario. Figure 4 visualizes the search process for EMFI targeting PIN verification on the 32-bit MCU. Similar to the VG scenario, the external offset parameter yielded mixed `normal` and `mute` verdicts across the entire range (Figure 4a). However, unlike VG, the offset parameter did not exhibit `mute` behavior in the specific range of 10–20. Regarding pulse intensity, while both verdict classes were observed across the voltage range, the proportion of `mute` outcomes increased at higher voltage levels. The spatial coordinates formed a boundary region in the center of the die. In this scenario as well, as shown in Figure 4b, our method successfully generated parameters along the boundary. Upon acquiring `intended fault` samples, the generation process dynamically adapted to prioritize the vicinity of these discovered samples, as illustrated in Figure 4c, while simultaneously maintaining exploration along the boundary. Figure 4d shows that the method successfully delineated the `intended fault` regions located on the boundary. These results also demonstrate that the `intended fault` distribution is nested within specific sub-regions of the boundary, consistent with VG scenario. Consequently, mere boundary exploration is insufficient.

A comparative analysis of the search coverage in the spatial domain (Figure 5) reveals trends consistent with the VG experiments: AZB prioritizes exploration, whereas GA, MA, SHA, SMAC, and CGAN favor exploitation. Specifically, while the RS results delineate a boundary region resembling a large inverted triangle, `intended faults` across all methods are concentrated within the coordinates ($90 < x < 150$ and $110 < y < 220$). This phenomenon corroborates our earlier analysis that `intended faults` are clustered in specific sub-regions. Our proposed method successfully balances these objectives, exploring the widest area of effective parameters while identifying a quantity of the `intended fault` parameters more than 2.6× greater than that of the state-of-the-art techniques.



(a) Initial random search results.

(b) Distribution generated by the ensemble trained on the initial dataset.



(c) Distribution after fifth fine-tuning iteration.

(d) Final cumulative fault results showing the discovered parameter landscape.

Figure 4: Visualization of the parameter search evolution for EMFI targeting PIN verification on the 32-bit MCU.

Quantitative Performance Summary. To quantitatively assess the efficacy of the proposed method in locating the `intended fault` distribution, we compared the number of distinct `intended fault` parameters. Figure 6 summarizes the average number of `intended fault` parameters discovered by each method under an identical injection budget. The experimental results demonstrate that our framework consistently outperforms all existing techniques, achieving significantly higher coverage in every scenario. Depending on the scenario, our method recovers between 1.2× and 42.2× more `intended fault` parameters than the best-performing baseline. The performance gap between our method and existing techniques was less pronounced in relatively easy fault injection scenarios, such as AES DFA. Conversely in challenging scenarios, such as PIN verification, our method demonstrated a significant advantage. The experimental results demonstrate that our method provides the most comprehensive characterization of the `intended fault` distribution.

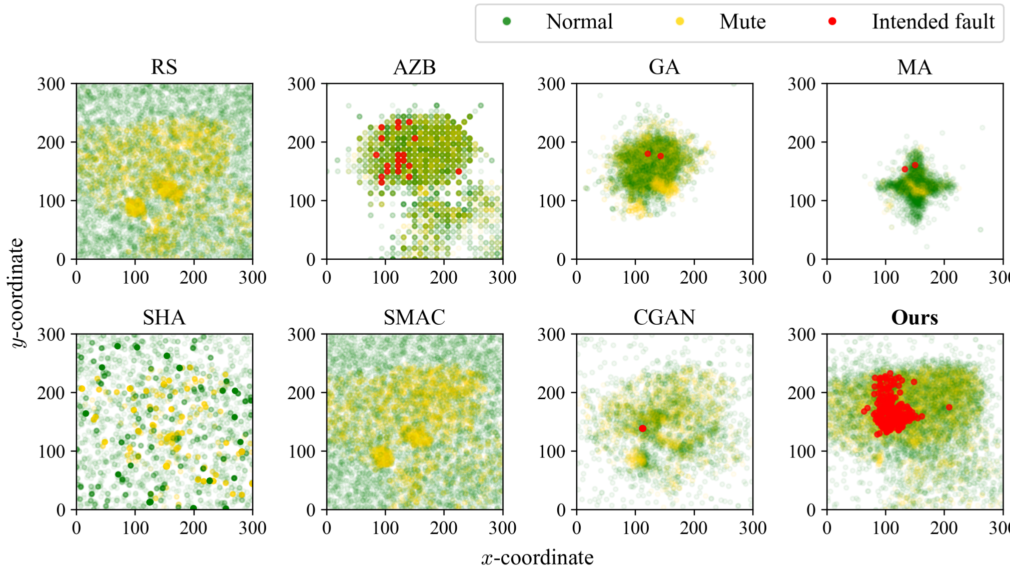


Figure 5: Comparison of candidate parameter distributions for EMFI (PIN, 32-bit MCU) projected onto the spatial plane.

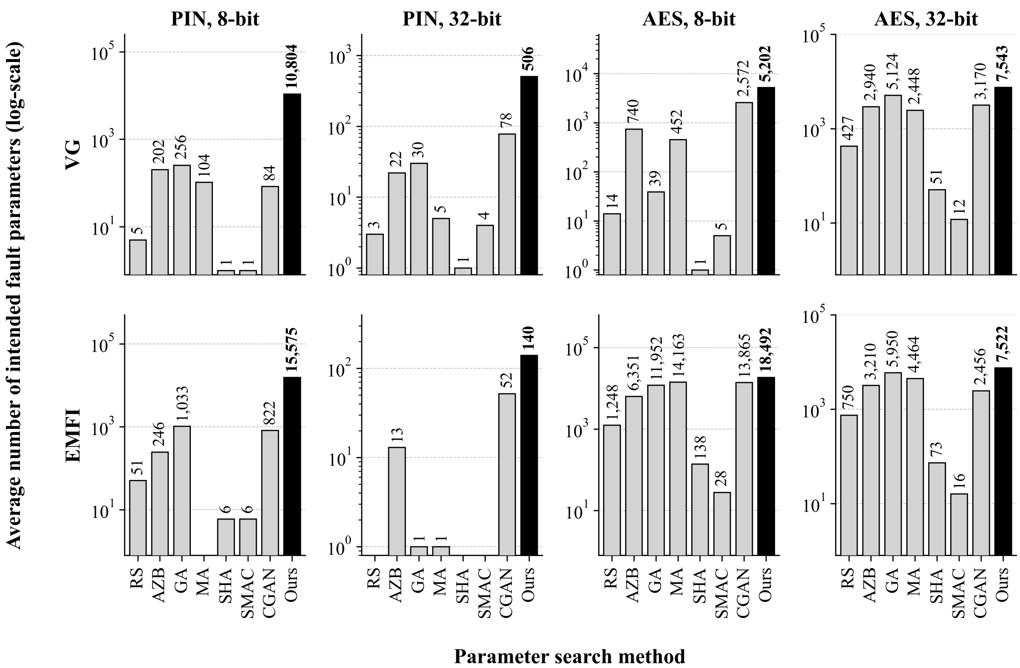


Figure 6: Quantitative comparison of the number of unique `intended fault` parameters identified by each method (log-scale).

4.3 Reliability of the Optimal Parameter

To assess the efficacy of the parameters identified by our framework, we conducted a validation phase focusing on reliability. Specifically, we refined the set of `intended fault` parameters discovered by each search strategy using RSHA to isolate the optimal parameter. Subsequently, we executed 3,000 independent validation injections for each optimal parameter to empirically quantify its reliability.

Figure 7 presents the comparative results. Our approach consistently achieved the highest success rates across all evaluated scenarios. This performance differential is particularly pronounced in challenging attack environments, such as the PIN verification on the 32-bit MCU. In this target, our approach yielded success rates exceeding the nearest state-of-the-art technique by 20.3 percentage points for VG and 15.8 percentage points for EMFI. Given that the validation sample size of 3,000 bounds the estimation error to within 2.34% at a 99% confidence level (based on Equation 2), the observed discrepancy in success rates signifies a statistically significant divergence in parameter reliability, rather than stochastic variance. These margins demonstrate that our method effectively characterizes the `intended fault` distribution, thereby facilitating the identification of highly robust parameter configurations.

Among the state-of-the-art techniques, AZB exhibited the highest success rates, attributed to its strong emphasis on exploration which allows for broad coverage of the parameter space. However, its lack of an exploitation mechanism prevented it from precisely refining the search within promising regions, resulting in suboptimal reliability compared to our method. Conversely, exploitation-oriented methods (GA, MA, SHA, SMAC, CGAN) frequently converged prematurely to local optima, yielding significantly lower success rates. This comparative analysis underscores the imperative of dynamically balancing exploration and exploitation to maximize the reliability of fault injection attacks.

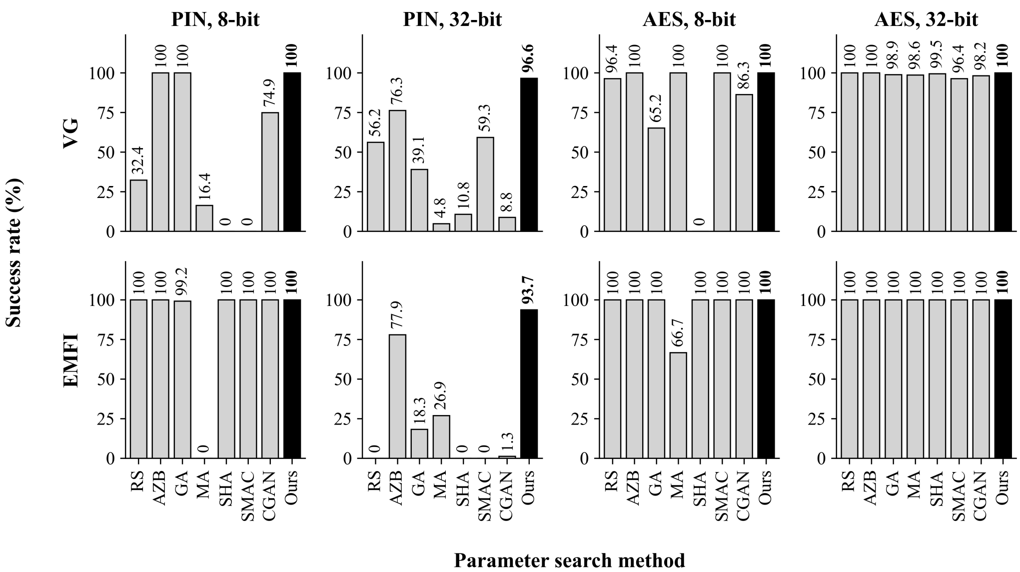


Figure 7: Evaluation of attack reliability: Success rates of the optimal parameters identified by each search strategy. The results are validated over 3,000 independent injections with the optimal parameter refined by RSHA.

5 Discussion

The fundamental premise underlying both this study and prior parameter search methods [CPB+13, PBJC14, PBBJ15, KBV24] is the *boundary hypothesis*. However, our empirical observations regarding a specific hardware configuration indicate that `intended faults` can manifest even in the absence of a distinct transition to the `mute` state. Specifically, in the EMFI experiments targeting the 8-bit MCU (as illustrated in Figure 8), we observed `intended fault` occurrences in regions characterized by a negligible frequency of `mute` verdicts, a behavior that significantly deviates from the standard boundary profile. A detailed analysis of the Random Search (RS) results reveals that `intended faults` form distinct clusters. The primary cluster, situated in the lower-left region ($150 < x < 250$ and $0 < y < 150$), exhibits a distribution comprising 3.38% `intended fault`, 18.94% `mute`, and 77.68% `normal` verdicts. This distribution aligns with the boundary hypothesis, characterized by a mixture of benign and severe outcomes. In contrast, the upper-right cluster ($200 < x < 330$ and $200 < y < 330$) presents a markedly different profile: 0.63% `intended fault`, 0.05% `mute`, and 99.32% `normal`. In this region, `mute` outcomes are scarcer than `intended faults`, and the overwhelming dominance of `normal` verdicts suggests that these faults manifest deep within the benign operational envelope rather than at a distinct transition zone. This finding implies the existence of `intended fault` regions that remain effectively invisible to strategies strictly adhering to boundary exploration.

To mitigate this limitation and achieve comprehensive coverage of the parameter space, one strategy is to include a subset of random parameters in each batch alongside those generated by our method, thereby enforcing a degree of global exploration. Furthermore, once `intended fault` parameters situated on indistinct boundaries are acquired, the deep learning model can learn from these instances to generate promising candidates in their vicinity. This adaptation is particularly valuable for analysts prioritizing exhaustive vulnerability assessment—such as identifying all spatially vulnerable locations on a chip—by shifting the trade-off balance in favor of exploration. However, it is crucial to note that the vast majority of `intended faults` observed in our experiments were concentrated within the boundary (e.g., the lower-left cluster). Consequently, if the adversarial objective is to efficiently identify a reliable fault injection parameter, diluting the optimization process with random samples may prove counterproductive.

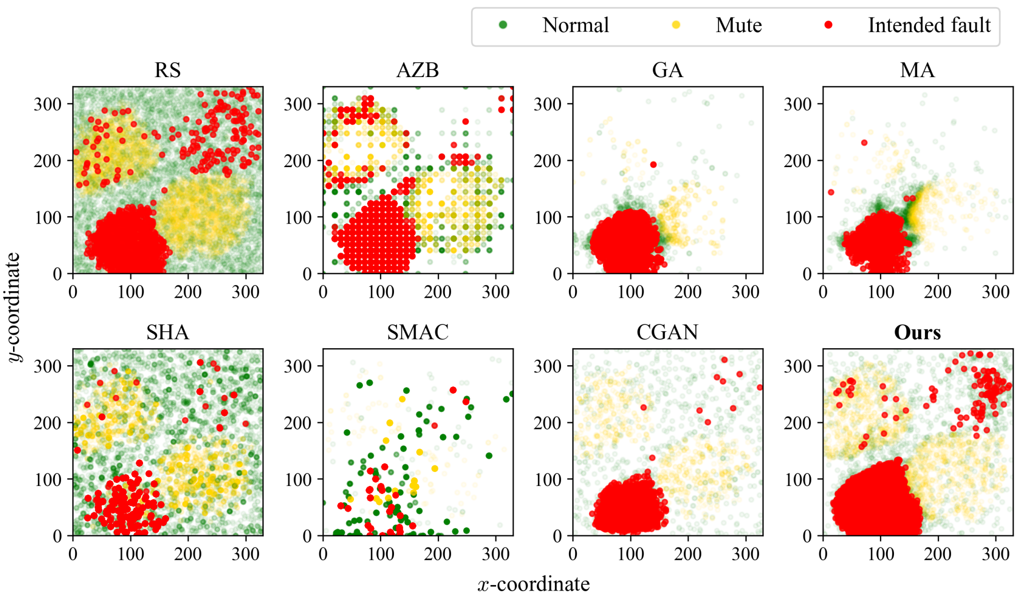


Figure 8: Parameter distributions for EMFI (AES, 8-bit MCU) projected onto the spatial plane. The plot highlights `intended fault` occurrences in regions lacking a distinct boundary with `mute` verdicts, specifically in the upper-right cluster.

6 Conclusion

In this paper, we presented a discriminative model-based framework for generating promising fault parameter candidates, designed to maximize success rates. We constructed an adaptive ensemble of two specialized models: a regression model for boundary exploration and a classification model for the exploitation of discovered `intended fault` samples. This framework effectively addresses the trade-off between exploration and exploitation that limits existing parameter search strategies. Furthermore, the introduction of the Refining Successive Halving Algorithm (RSHA) enables the rigorous sieving of promising candidates, ensuring the efficient identification of the global optimum among the discovered `intended fault` parameters with statistical guarantees.

We validated the proposed framework through extensive experiments across eight diverse scenarios involving Voltage Glitching (VG) and Electromagnetic Fault Injection (EMFI). The results confirm that our method provides a more comprehensive characterization of the `intended fault` distribution compared to state-of-the-art techniques. Specifically, our approach discovered up to 42.2× more `intended fault` parameters and achieved success rate improvements of up to 20.3 percentage points. Consequently, our approach facilitates the induction of `intended faults` with significantly higher reliability than existing methods.

Additionally, we empirically observed that in a specific hardware configuration, `intended faults` can manifest in regions where the boundary is indistinct and difficult to recognize, characterized by a negligible frequency of `mute` verdicts. Although the majority of exploitable parameters were still concentrated at distinct boundaries, this finding implies that the conventional boundary hypothesis may be insufficient for locating all possible fault regions. Therefore, for analysts prioritizing the exhaustive mapping of the entire `intended fault` parameter space, relying solely on boundary-centric heuristics may be inadequate. We discussed strategic adaptations to address this limitation. Future work will focus on developing methodologies to efficiently explore these elusive fault regions.

References

[AK97] R. Anderson and M. Kuhn. Low cost attacks on tamper resistant devices. In *5th Security Protocols Workshop*, volume 1361 of *LNCS*, pages 125–136. Springer, April 1997.

[BCV13] Yoshua Bengio, Aaron C. Courville, and Pascal Vincent. Representation learning: A review and new perspectives. *IEEE Trans. Pattern Anal. Mach. Intell.*, 35(8):1798–1828, 2013. doi:10.1109/TPAMI.2013.50.

[BFP19] Claudio Bozzato, Riccardo Focardi, and Francesco Palmarini. Shaping the glitch: Optimizing voltage fault injection attacks. *IACR TCHES*, 2019(2):199–224, 2019. URL: https://tches.iacr.org/index.php/TCHES/article/view/7390, doi:10.13154/tches.v2019.i2.199-224.

[Bis07] Christopher M. Bishop. *Pattern recognition and machine learning, 5th Edition*. Information science and statistics. Springer, 2007. URL: https://www.worldcat.org/oclc/71008143.

[Bri50] Glenn W. Brier. Verification of Forecasts Expressed in Terms of Probability. *Monthly Weather Review*, 78(1):1, January 1950. doi:10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.

[BS97] Eli Biham and Adi Shamir. Differential fault analysis of secret key cryptosystems. In Burton S. Kaliski, Jr., editor, *CRYPTO'97*, volume 1294 of *LNCS*, pages 513–525. Springer, Berlin, Heidelberg, August 1997. doi:10.1007/BFb0052259.

[CBHK02] Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: synthetic minority over-sampling technique. *J. Artif. Intell. Res.*, 16:321–357, 2002. URL: https://doi.org/10.1613/jair.953, doi:10.1613/JAIR.953.

[CPB+13] Rafael Boix Carpi, Stjepan Picek, Lejla Batina, Federico Menarini, Domagoj Jakobovic, and Marin Golub. Glitch it if you can: Parameter search strategies for successful fault injection. In Aurélien Francillon and Pankaj Rohatgi, editors, *Smart Card Research and Advanced Applications - 12th International Conference, CARDIS 2013, Berlin, Germany, November 27-29, 2013. Revised Selected Papers*, volume 8419 of *Lecture Notes in Computer Science*, pages 236–252. Springer, 2013. doi:10.1007/978-3-319-08302-5_16.

[GPM+14] Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron C. Courville, and Yoshua Bengio. Generative adversarial nets. In Zoubin Ghahramani, Max Welling, Corinna Cortes, Neil D. Lawrence, and Kilian Q. Weinberger, editors, *Advances in Neural Information Processing Systems 27: Annual Conference on Neural Information Processing Systems 2014, December 8-13 2014, Montreal, Quebec, Canada*, pages 2672–2680, 2014. URL: https://proceedings.neurips.cc/paper/2014/hash/5ca3e9b122f61f8f06494c97b1afccf3-Abstract.html.

[GZW+25] Xue Gong, Xin Zhang, Qianmei Wu, Fan Zhang, Junge Xu, Qingni Shen, and Zhi Zhang. Practical opcode-based fault attack on AES-NI. *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, 2025(3):693–716, 2025. URL: https://doi.org/10.46586/tches.v2025.i3.693-716, doi:10.46586/TCHES.V2025.I3.693-716.

[HYS16] Le Hou, Chen-Ping Yu, and Dimitris Samaras. Squared earth mover's distance-based loss for training deep neural networks. *CoRR*, abs/1611.05916, 2016. URL: http://arxiv.org/abs/1611.05916, arXiv:1611.05916.

[KBV24] Troya Çagil Köylü, Cornelis Christiaan Berg, and Praveen Kumar Vadnala. Cgan-based automated fault injection. In *IEEE European Test Symposium, ETS 2024, The Hague, Netherlands, May 20-24, 2024*, pages 1–6. IEEE, 2024. doi:10.1109/ETS61313.2024.10568027.

[KFP21] Marina Krcek, Daniele Fronte, and Stjepan Picek. On the importance of initial solutions selection in fault injection. In *18th Workshop on Fault Detection and Tolerance in Cryptography, FDTC 2021, Milan, Italy, September 17, 2021*, pages 1–12. IEEE, 2021. doi:10.1109/FDTC53659.2021.00011.

[KGAS25] Anup Kumar Kundu, Shibam Ghosh, Aikata Aikata, and Dhiman Saha. Tofa: Towards fault analysis of GIFT and gift-like ciphers leveraging truncated impossible differentials. *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, 2025(3):614–643, 2025. URL: https://doi.org/10.46586/tches.v2025.i3.614-643, doi:10.46586/TCHES.V2025.I3.614-643.

[KK99] Oliver Kömmerling and Markus G. Kuhn. Design principles for tamper-resistant smartcard processors. In Scott B. Guthery and Peter Honeyman, editors, *Proceedings of the 1st Workshop on Smartcard Technology, Smartcard 1999, Chicago, Illinois, USA, May 10-11, 1999*. USENIX Association, 1999. URL: https://www.usenix.org/conference/usenix-workshop-smartcard-technology/design-principles-tamper-resistant-smartcard.

[KOFP22] Marina Krcek, Thomas Ordas, Daniele Fronte, and Stjepan Picek. The more you know: Improving laser fault injection with prior knowledge. In *Workshop on Fault Detection and Tolerance in Cryptography, FDTC 2022, Virtual Event / Italy, September 16, 2022*, pages 18–29. IEEE, 2022. doi:10.1109/FDTC57191.2022.00012.

[MSPB18] Antun Maldini, Niels Samwel, Stjepan Picek, and Lejla Batina. Genetic algorithm-based electromagnetic fault injection. In *2018 Workshop on Fault Diagnosis and Tolerance in Cryptography, FDTC 2018, Amsterdam, The Netherlands, September 13, 2018*, pages 35–42. IEEE Computer Society, 2018. doi:10.1109/FDTC.2018.00014.

[Newa] NewAE Technology Inc. ChipWhisperer-Husky. https://chipwhisperer.readthedocs.io/en/latest/Capture/ChipWhisperer-Husky.html.

[Newb] NewAE Technology Inc. CW303 XMEGA Target. https://chipwhisperer.readthedocs.io/en/latest/Targets/CW303%20XMEGA.html.

[Newc] NewAE Technology Inc. CW312T-SAM4S. https://chipwhisperer.readthedocs.io/en/latest/chipwhisperer-target-cw308t/CW312T_ATSAM4S/README.html.

[Newd] NewAE Technology Inc. CW520 ChipSHOUTER. https://chipwhisperer.readthedocs.io/en/latest/ChipSHOUTER/ChipSHOUTER.html.

[NJ01] Andrew Y. Ng and Michael I. Jordan. On discriminative vs. generative classifiers: A comparison of logistic regression and naive bayes. In Thomas G. Dietterich, Suzanna Becker, and Zoubin Ghahramani, editors, *Advances in Neural Information Processing Systems 14 [Neural Information Processing Systems: Natural and Synthetic, NIPS 2001, December 3-8, 2001, Vancouver, British Columbia, Canada]*, pages 841–848. MIT Press, 2001. URL: https://proceedings.neurips.cc/paper/2001/hash/7b7a53e239400a13bd6be6c91c4f6c4e-Abstract.html.

[PBBJ15] Stjepan Picek, Lejla Batina, Pieter Buzing, and Domagoj Jakobovic. Fault injection with a new flavor: Memetic algorithms make a difference. In Stefan Mangard and Axel Y. Poschmann, editors, *Constructive Side-Channel Analysis and Secure Design - 6th International Workshop, COSADE 2015, Berlin, Germany, April 13-14, 2015. Revised Selected Papers*, volume 9064 of *Lecture Notes in Computer Science*, pages 159–173. Springer, 2015. doi:10.1007/978-3-319-21476-4_11.

[PBJC14] Stjepan Picek, Lejla Batina, Domagoj Jakobovic, and Rafael Boix Carpi. Evolving genetic algorithms for fault injection attacks. In *37th International Convention on Information and Communication Technology, Electronics and Microelectronics, MIPRO 2014, Opatija, Croatia, May 26-30, 2014*, pages 1106–1111. IEEE, 2014. doi:10.1109/MIPRO.2014.6859734.

[PQ03] Gilles Piret and Jean-Jacques Quisquater. A differential fault attack technique against SPN structures, with application to the AES and KHAZAD. In Colin D. Walter, Çetin Kaya Koç, and Christof Paar, editors, *CHES 2003*, volume 2779 of *LNCS*, pages 77–88. Springer, Berlin, Heidelberg, September 2003. doi:10.1007/978-3-540-45238-6_7.

[SA03] Sergei P. Skorobogatov and Ross J. Anderson. Optical fault induction attacks. In Burton S. Kaliski, Jr., Çetin Kaya Koç, and Christof Paar, editors, *CHES 2002*, volume 2523 of *LNCS*, pages 2–12. Springer, Berlin, Heidelberg, August 2003. doi:10.1007/3-540-36400-5_2.

[SAW25] Kevin Schneider, Lukas Auer, and Alexander Wagner. Fault attacks on ECC signature verification. *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, 2025(4):1010–1052, 2025. URL: https://doi.org/10.46586/tches.v2025.i4.1010-1052, doi:10.46586/TCHES.V2025.I4.1010-1052.

[SC22] Divya Saxena and Jiannong Cao. Generative adversarial networks (gans): Challenges, solutions, and future directions. *ACM Comput. Surv.*, 54(3):63:1–63:42, 2022. doi:10.1145/3446374.

[VOGT21] Jan Van den Herrewegen, David Oswald, Flavio D. Garcia, and Qais Temeiza. Fill your boots: Enhanced embedded bootloader exploits via fault injection and binary analysis. *IACR TCHES*, 2021(1):56–81, 2021. URL: https://tches.iacr.org/index.php/TCHES/article/view/8727, doi:10.46586/tches.v2021.i1.56-81.

[WMP21] Vincent Werner, Laurent Maingault, and Marie-Laure Potet. Fast calibration of fault injection equipment with hyperparameter optimization techniques. In Vincent Grosso and Thomas Pöppelmann, editors, *Smart Card Research and Advanced Applications - 20th International Conference, CARDIS 2021, Lübeck, Germany, November 11-12, 2021, Revised Selected Papers*, volume 13173 of *Lecture Notes in Computer Science*, pages 121–138. Springer, 2021. doi:10.1007/978-3-030-97348-3_7.

[WMT25] Weizhe Wang, Pierrick Méaux, and Deng Tang. Shortcut2secrets: A table-based differential fault attack framework. *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, 2025(2):385–419, 2025. URL: https://doi.org/10.46586/tches.v2025.i2.385-419, doi:10.46586/TCHES.V2025.I2.385-419.